For example, in the category of modules over a ring A, the free module A is a projective generator.

Exercise 9.6.3. Show that in a finite abelian category, P is a projective generator if and only if for every simple object L, one has $\mathrm{Hom}(P, L) \neq 0$. Deduce that any finite abelian category has a projective generator.

Now let $\mathcal{C}$ be a finite abelian category with a projective generator P. Let $B = \mathrm{End}(P)^{\mathrm{op}}$ (a finite dimensional algebra acting on the right on P). Let $B - \mathrm{fmod}$ denote the category of finite dimensional left B-modules. Consider the functor $F : \mathcal{C} \to B - \mathrm{fmod}$ given by the formula $F(M) = \mathrm{Hom}(P, M)$.

Theorem 9.6.4. *The functor F is an equivalence of categories. Thus, any finite abelian category over a field k is equivalent to the category of modules over a finite dimensional k-algebra.*

The proof of Theorem 9.6.4 is contained in the following problem.

Problem 9.6.5. Let $G : B - \mathrm{fmod} \to \mathcal{C}$ be the functor defined by $G(X) := P \otimes_B X$, where $P \otimes_B X$ is the cokernel of the morphism $\psi : P \otimes B \otimes X \to P \otimes X$ given by $\psi = a_P \otimes \mathrm{Id} - \mathrm{Id} \otimes a_X$ (where $a_P : P \otimes B \to P, a_X : B \otimes X \to X$ are the morphisms representing the actions of B on P and X).

(i) Show that $F \circ G \cong \mathrm{Id}$. That is, for every $X \in B - \mathrm{fmod}$, show that X is naturally isomorphic to $\mathrm{Hom}(P, P \otimes_B X)$. (For this you should only need that P is a nonzero projective object.)

(ii) For any $X \in \mathcal{C}$, construct a natural morphism

$$\xi : P \otimes_B \mathrm{Hom}(P, X) \to X,$$

and show that it is surjective.

(iii) Show that $G \circ F \cong \mathrm{Id}$. To this end, consider the short exact sequence

$$0 \to K \to P \otimes_B \mathrm{Hom}(P, X) \to X \to 0,$$

where the third map is ξ. Apply the functor F to this sequence and use (i) to conclude that $K = 0$ and hence ξ is an isomorphism. Conclude that the functors G and F are quasi-inverse to each other and hence are equivalences of categories.